Promo Finder: solution

1. What it does

Reads timestamped screenplays and returns ranked 20–60 s promos (segments, hook, cliffhanger, rationale, score, publishing plan) plus every rejected candidate with the reason. It only selects moments.

2. How it works (five stages)

1. **Parse**: scripts become numbered blocks (dialogue, ACTION, SFX, CAMERA) with source times (3,727 blocks).
2. **Scout (LLM)**: proposes clips per episode, plus cross-episode ideas planned from the loglines. **It picks block numbers, never timestamps**; times are copied from the script, so they cannot be hallucinated. It can still misdescribe what it picked, hence stages 3–5.
3. **Validate (code)**: 20–60 s; hook starts in the first 3 s; clip ends ≤ 4 s after the cliffhanger; hook and cliffhanger supported by the selected screenplay blocks (inside the cut, not in a trimmed shot); no dialogue cut mid-line; segments in story order. A failing clip is first **repaired by trimming its edges in code**, then gets one LLM repair attempt; every change is logged.
4. **Blind critic (LLM)**: a second prompt sees only the clip, as a first-time viewer would, and scores the 100-point rubric. It flags weak hook, missing context, predictable ending, low curiosity and misleading edit, and its scores replace the scout's; clear failures are rejected. **One weak subjective score on a clip totalling 75 or more goes to human review instead of rejection.** Reviews of identical clips are reused from the cache.
5. **Rank and review**: dedups, ranks, writes a publishing plan and runs local **human-review checks**. An **independent verifier** (own parser) then re-checks every timestamp, duration and score against the raw scripts.

- *Criteria:* critic scores for hook (25), standalone clarity (20), cliffhanger (25), emotion (15) and coherence (10), plus duration and boundary quality from code (5) = 100.

- *Mechanism (code, not LLM):* clips without a borderline score first, then total score, stronger hook, shorter clip; of two overlapping clips the unflagged one stays. The verifier fails any output out of this order.

- *Publishing order:* the rank (1–5 High, 6–10 Medium, rest Low). One LLM call explains each rank and writes a plan weighing hook strength, context needed, spoiler risk (early-story first) and arc variety; it cannot reorder.

3. Validation and the human in the loop

Free LLMs are imperfect, so the tool states its uncertainty. Each review flag names its evidence and suggests **accept / inspect / repair / reject**; subjective concerns never auto-reject, and an editor decides.

Kind	Checks	Basis
Objective	hook late or outside the cut; cliffhanger outside it; camera-only hook; split sentence; story order; displayed vs source times; missing or zero critic score	exact, from blocks
Approximate	hook/cliffhanger description not supported by the clip	word overlap; asks for a look
Subjective	critic flags; low coherence; scout vs critic disagree by ≥ 15 points	LLM judgement

`audit` re-checks completed runs, while `compare` evaluates different runs with the same method; both make **no model calls**. Six known good/bad clips check the critic (v3 reuses earlier scores for them). 62 tests cover broken clips (weak hooks, misleading endings, missing context, mid-line cuts), tampered outputs, quota errors and cache accounting.

4. Results and the recall study

The raw output holds v3, the final run. v2 (first version) and v4 (an experiment) were development runs, shown only for comparison: same models (free Gemini, batch mode) and match rule (a promo covers ≥ 50 % of a reference).

	v2 (first version)	**v3 (final, in raw output)**	**v4 (experiment)**
Promos: recommended / needs review	31: 14 / 17	48: **17 / 31**	54: 18 / 36
Reference recall, all promos	8 / 17 (47 %)	**13 / 17 (76 %)**	12 / 17 (71 %)
Reference recall, recommended only	5 / 17	5 / 17	4 / 17
Critic probes correct	6 / 6*	6 / 6	6 / 6

Why v2 missed 9 references: all 9 *had been proposed*, then lost: the LLM repair returned the same broken range (R02, R13), one low subjective score rejected an 81–82/100 clip (R07, R05), or a cross-episode setup was never paired (R08). **v3** fixes 27 of 30 invalid clips by trimming in code and sends one-weakness clips to review instead of rejecting them. Recall over all promos rose 47 % → 76 %: of the 13 matched references, 5 are among recommended promos and 8 are flagged for review, so **76 % is discovery recall, not publication-ready recall**. Of 19 new promos, 9 are good, 5 borderline, 5 poor (quality-adjusted gain +3, not +5). **v3 is the final result:** the clean list grew (14 → 17) and weaker additions are labelled (`RECALL_STUDY.md` in the raw output).

v4 (re-scouting episodes without a clip, arc planner v2) added 7 promos and 5 episodes but no recall (with a dedup fix it equals v3) and missed R08 again, so it was not adopted. *Probes and the 75-point rule were set after seeing data, and the references were picked by an AI, so recall is a consistency check, not ground truth.

5. Assumptions, models and scale

Assumptions: script times match the video; cuts fall on block boundaries (shots may be trimmed, lines never); a new viewer knows only the clip; spoilers affect order, not score. **Models:** one `complete(system, prompt)` interface: Claude (default), Gemini, Groq, OpenRouter, Ollama, any OpenAI-compatible API, cache-only replay, or an offline heuristic; switching is one flag. For free tiers it paces requests, retries on HTTP 429, stops on the daily quota and resumes from its cache. `--batch` puts 10 episodes or 15 clips in a request (about 16 requests instead of 250).

6. Why results are limited, and how to improve them

Discovery recall is 76 %, but **only 5 / 17 references are among the recommended clips**. v3's own output shows why:

- **The free models are imprecise.** 30 of 82 proposed clips broke a timing rule (code fixed 27), and 7 promos carry a hook or cliffhanger description their clip does not support.
- **Their judgement is noisy.** Of 31 promos sent to review, 15 have scout and critic scores ≥ 15 points apart and 11 one weak subjective score; 5 have both, so 21 carry either flag and 10 were flagged for other reasons. Batching 10 episodes or 15 clips per request also makes reviews less independent.
- **Arcs are planned from loglines**, so long-range pay-offs (R08) are missed; the 17 AI-picked references are a small, subjective yardstick; and text cannot show acting or pace.

A stronger model (e.g. Claude Opus 5.5, `--provider anthropic`, no code change) targets the model errors: more exact block choices, accurate descriptions, steadier scores and, with a paid quota, unbatched requests. It cannot fix the text-only limit or the subjective references; **not yet measured** (run it, then `compare` with v3). **Other levers:** editor-labelled references and probes to set thresholds; `--per-episode 5`; two critics that must agree; arc planning from full scripts; a rendered-video check.

7. AI tools used, and where AI got it wrong

Claude Code (Claude Opus 5.5) wrote the code, tests, reference picks and docs; free Gemini runs the pipeline. Mistakes caught, all fixed and tested: **the independent verifier caught an AI-written validator** letting a promo end 0.2 s into a line of dialogue; **the audit found five promos whose cliffhanger block fell outside the cut** (the validator counted trimmed shots); and **Gemini wrote `"narrative_ coherence"`**, read as 0/10 for P029, so keys are now normalised.